

Article

The Effects of Charged Amino Acid Side-Chain Length on Diagonal Cross-Strand Interactions between Carboxylate- and Ammonium-Containing Residues in a β -Hairpin

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Abstract: The β -sheet is one of the common protein secondary structures, and the aberrant aggregation of β -sheets is implicated in various neurodegenerative diseases. Cross-strand interactions are an important determinant of β -sheet stability. Accordingly, both diagonal and lateral cross-strand interactions have been studied. Surprisingly, diagonal cross-strand ion-pairing interactions have yet to be investigated. Herein, we present a systematic study on the effects of charged amino acid side-chain length on a diagonal ion-pairing interaction between carboxylate- and ammonium-containing residues in a β -hairpin. To this end, 2D-NMR was used to investigate the conformation of the peptides. The fraction folded population and the folding free energy were derived from the chemical shift data. The fraction folded population for these peptides with potential diagonal ion pairs was mostly lower compared to the corresponding peptide with a potential lateral ion pair. The diagonal ion-pairing interaction energy was derived using double mutant cycle analysis. The Asp2-Dab9 (Asp: one methylene; Dab: two methylenes) interaction was the most stabilizing (-0.79 ± 0.14 kcal/mol), most likely representing an optimal balance between the entropic penalty to enable the ion-pairing interaction and the number of side-chain conformations that can accommodate the interaction. These results should be useful for designing β -sheet containing molecular entities for various applications.

Keywords: peptide; β -hairpin; diagonal interaction; ion-pairing interaction; charged amino acid; side-chain length

1. Introduction

The β -sheet is one of the common protein secondary structures, with 23% of the protein residues adopting a β -sheet conformation [1–3]. Aberrant aggregation of β -sheets form amyloid fibrils involved in a number of neurodegenerative diseases including Alzheimer's disease [4,5], Huntington's disease [6,7], and Parkinson's disease [8]. As such, studies on β -sheet stability and formation may provide insight into the emergence and treatment of these neurodegenerative diseases.

β -Sheet stability is determined by both the thermodynamic sheet propensity of the constituting residues [9,10] and cross-strand interactions [11–13]. A survey focusing on antiparallel β -sheets highlighted the importance of lateral cross-strand interactions between oppositely charged residues [11]. Accordingly, lateral cross-strand interactions have been studied in various sheet-containing host systems, including the protein G B1 domain [14,15], a zinc finger domain [16], and β -hairpins [17–27]. In the protein G B1 domain system, both lateral Glu-Lys and the Glu-Arg interactions provided approximately 1.0 kcal/mol stabilization based on thermal denaturation experiments [14]. In the zinc-finger system,

Article

帶電氨基酸側鏈長度對 β -髮夾中羧酸根和含鉍殘基對角交叉鏈交互作用的影響

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摘要: β -折疊是常見的蛋白質二級結構之一, β -折疊的異常聚集與多種神經退化性疾病有關。跨鏈交互作用是 β -片層穩定性的重要決定因素。因此, 對角線和橫向交叉鏈相互作用都得到了研究。令人驚訝的是, 對角跨鏈離子對相互作用尚未被研究。在此, 我們提出了一項系統性研究, 研究帶電氨基酸側鏈長度對 β -髮夾中羧酸根和含鉍殘基之間的對角離子對相互作用的影響。為此, 使用二維核磁共振來研究勝肽的構象。折疊布居分數和折疊自由能來自化學位移數據。與具有潛在橫向離子對的相應勝肽相比, 這些具有潛在對角離子對的勝肽的折疊群體分數大多較低。使用雙突變體循環分析得出對角離子對相互作用能。Asp2-Dab9 (Asp: 一個亞甲基; Dab: 兩個亞甲基) 相互作用是最穩定的 (-0.79 ± 0.14 kcal/mol), 很可能代表了實現離子對相互作用的熵損失與可以容納相互作用的側鏈構象數量之間的最佳平衡。這些結果對於設計包含用於各種應用的分子實體的 β -片應該有用。

關鍵字: 勝肽; β -髮夾; 對角相互作用; 離子對相互作用; 帶電氨基酸; 側鏈長度

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一、簡介

β -摺疊是常見的蛋白質二級結構之一, 23%的蛋白質殘基採 β -摺疊構象[1-3]。 β -片層的異常聚集形成澱粉樣原纖維, 與許多神經退化性疾病有關, 包括阿茲海默症 [4,5]、亨廷頓舞蹈症 [6,7] 和帕金森氏症 [8]。因此, 對 β -片穩定性和形成的研究可能有助於深入了解這些神經退化性疾病的出現和治療。

β -片層穩定性由構成殘基的熱力學片層傾向[9,10]和跨鏈交互作用[11-13]決定。一項針對反平行 β -片的調查強調了帶相反電荷的殘基之間橫向跨鏈相互作用的重要性 [11]。因此, 橫向跨鏈相互作用已在各種含片層的宿主系統中進行了研究, 包括蛋白質 G B1 結構域 [14,15]、鋅指結構域 [16] 和 β -髮夾 [17-27]。在蛋白質 G B1 結構域系統中, 基於熱變性實驗, 橫向 Glu-Lys 和 Glu-Arg 相互作用提供了約 1.0 kcal/mol 的穩定性 [14]。在鋅指系統中,

the lateral cross-strand interaction between negatively charged residues (Asp and Glu) and positively charged residues (Arg and Lys) were 0.02–0.48 kcal/mol based on competitive metal ion binding studies [16]. In β -hairpins, lateral cross-strand Glu–Lys ion-pairing interactions provided 0.1–0.3 kcal/mol stabilization based on NMR studies [20,22,23]. Furthermore, various types of lateral cross-strand interactions have been investigated in β -hairpins, including hydrophobics [17,24], electrostatics [17,20,22,25–27], and aromatic π – π interactions [21,22]. Nonetheless, statistical studies showed that diagonal cross-strand interactions are much more significant, compared to lateral cross-strand interactions across antiparallel β -strands [12].

Diagonal cross-strand interactions are enabled by the inherent right-handed twist of β -sheet structures [19,27,28]. Accordingly, a number of diagonal cross-strand interactions have been investigated in β -hairpins, including hydrophobics [24], cation– π [29–32], carbohydrate– π [33,34], and amide– π interactions [35]. These diagonal cross-strand interactions stabilized the β -hairpin structure by 0.1–1.0 kcal/mol [24,29–35]. Surprisingly, the diagonal cross-strand interactions between oppositely charged residues (i.e., ion-pairing interactions) have yet to be investigated. Herein, we present a systematic study on the effects of charged amino acid side-chain length on diagonal cross-strand interactions between carboxylate- and ammonium-containing residues in a β -hairpin.

2. Results

2.1. Peptide Design and Synthesis

The experimental HPDZbbXaa peptides were designed based on Gellman's YKL peptide [19] and the HPTZbbXaa hairpin peptides described in our previous studies (Figure 1) [3,25,26]. A diagonal cross-strand cation– π interaction was observed in the parent YKL β -hairpin peptide [19]. Furthermore, diagonal cross-strand NOEs were observed between positions 2 and 9 in the HPTZbbXaa peptides [3,25–27]. Therefore, the Tyr2 (in peptide YKL) was replaced with negatively charged carboxylate-containing residues to interact with the positively charged ammonium-containing residues at position 9. The Glu4 (in peptide YKL) was replaced with Thr to maintain the β -structure based on the high thermodynamic sheet propensity of Thr [9,10] and to remove the potential lateral ion-pairing interaction with position 9. An acetyl group and a carboxamide group were incorporated at the N- and C-termini, respectively, to minimize unexpected interactions involving the charged termini [17]. To investigate the effects of charged amino acid side-chain length on diagonal cross-strand ion-pairing interactions, the negatively charged carboxylate-containing residues (Zbb = Asp, Glu, Aad) and positively charged ammonium-containing residues (Xaa = Dap, Dab, Orn, Lys) were incorporated at positions 2 and 9, respectively, to give the experimental HPDZbbXaa peptides (Figure 1). The experimental peptides were named HPDZbbXaa, representing a hairpin peptide to study the diagonal interactions, followed by the negatively charged Zbb residue at position 2 and the positively charged Xaa residue at position 9.

The fully folded reference peptides HPDFZbbXaa and the fully unfolded reference peptides HPDUZbbXaa were required to determine the fraction folded population of the experimental HPDZbbXaa peptides (Figure 1b) [3,19,25–27,36–38]. The fully folded reference HPDFZbbXaa peptides were designed by introducing cysteines at both the N- and C-termini of experimental HPDZbbXaa peptides to form intramolecular disulfide bonds [3,19,25–27,36–38]. The fully unfolded reference HPDUZbbXaa peptides were designed by replacing DPro with LPro at position 6 of the experimental HPDZbbXaa peptides, to disrupt the β -hairpin conformation [3,19,25–27,36–38].

根據競爭性金屬離子結合研究，帶負電的殘基（Asp 和 Glu）與帶正電的殘基（Arg 和 Lys）之間的橫向跨鏈相互作用為 0.02–0.48 kcal/mol [16]。根據 NMR 研究，在 β -髮夾中，橫向跨鏈 Glu-Lys 離子對相互作用提供了 0.1–0.3 kcal/mol 的穩定性 [20,22,23]。此外，人們也研究了 β -髮夾中各種類型的橫向跨鏈交互作用，包括疏水性[17,24]、靜電性[17,20,22,25–27]和芳香族 π - π 交互作用[21,22]。儘管如此，統計研究表明，與反平行 β 鏈之間的橫向跨鏈交互作用相比，對角線跨鏈交互作用要重要得多 [12]。

對角跨鏈相互作用是透過 β -片結構固有的右手扭曲實現的[19,27,28]。因此，在 β -髮夾中研究了許多對角線跨鏈相互作用，包括疏水性[24]、陽離子- π [29–32]、碳水化合物- π [33,34]和鹽胺- π 交互作用[35]。這些對角跨鏈交互作用使 β -髮夾結構穩定 0.1–1.0 kcal/mol [24,29–35]。令人驚訝的是，帶相反電荷的殘基之間的對角線跨鏈相互作用（即離子對相互作用）尚未被研究。在此，我們提出了一項系統性研究，研究帶電胺基酸側鏈長度對 β -髮夾中含羧酸根和含鈹殘基之間對角跨鏈相互作用的影響。

2. 結果

2.1. Peptide Design and Synthesis

實驗性 HPDZbbXaa 勝肽是根據 Gellman 的 YKL 勝肽 [19] 和我們先前的研究中所描述的 HPTZbbXaa 髮夾勝肽設計的（圖 1）[3,25,26]。在母體 YKL β -髮夾勝肽中觀察到對角交叉鏈陽離子- π 相互作用 [19]。此外，在 HPTZbbXaa 勝肽的位置 2 和 9 之間觀察到對角交叉鏈 NOE [3,25–27]。因此，Tyr2（肽 YKL 中）被帶負電荷的含羧酸殘基取代，與位置 9 處帶正電荷的含鈹殘基相互作用。Glu4（勝肽 YKL 中）被 Thr 取代，以維持基於 Thr [9,10] 的高熱力學片傾向的 β 結構，並消除與位置 9 潛在的橫向離子對相互作用。乙鹽基和甲鹽胺基分別是 N 端和 C 端，以盡量減少涉及帶電端的意外交互作用 [17]。為了研究帶負電的胺基酸側鏈長度對對角交叉鏈離子對相互作用的影響，將帶負電的含羧酸殘基（Zbb = Asp、Glu、Aad）和帶正電的含鈹殘基（Xaa = Dap、Dab、Orn、Lys）分別摻入位置 29，得到實驗性勝肽和 9，實驗勝肽被命名為 HPDZbbXaa，代表一個用於研究對角線交互作用的髮夾勝肽，後面是第 2 位帶負電的 Zbb 殘基和第 9 位帶正電的 Xaa 殘基。

需要完全折疊的參考肽 HPDFZbbXaa 和完全展開的參考肽 HPDUZbbXaa 來確定實驗 HPDZbbXaa 肽的折疊群體分數（圖 1b）[3,19,25–27,36–38]。完全折疊的參考 HPDFZbbXaa 肽是透過在實驗 HPDZbbXaa 肽的 N 端和 C 端引入半胱氨酸以形成分子內二硫鍵而設計的 [3,19,25–27,36–38]。完全展開的參考 HPDUZbbXaa 勝肽是透過用 LPro 取代實驗 HPDZbbXaa 勝肽第 6 位的 DPro 來設計的，以破壞 β -髮夾構象 [3,19,25–27,36–38]。

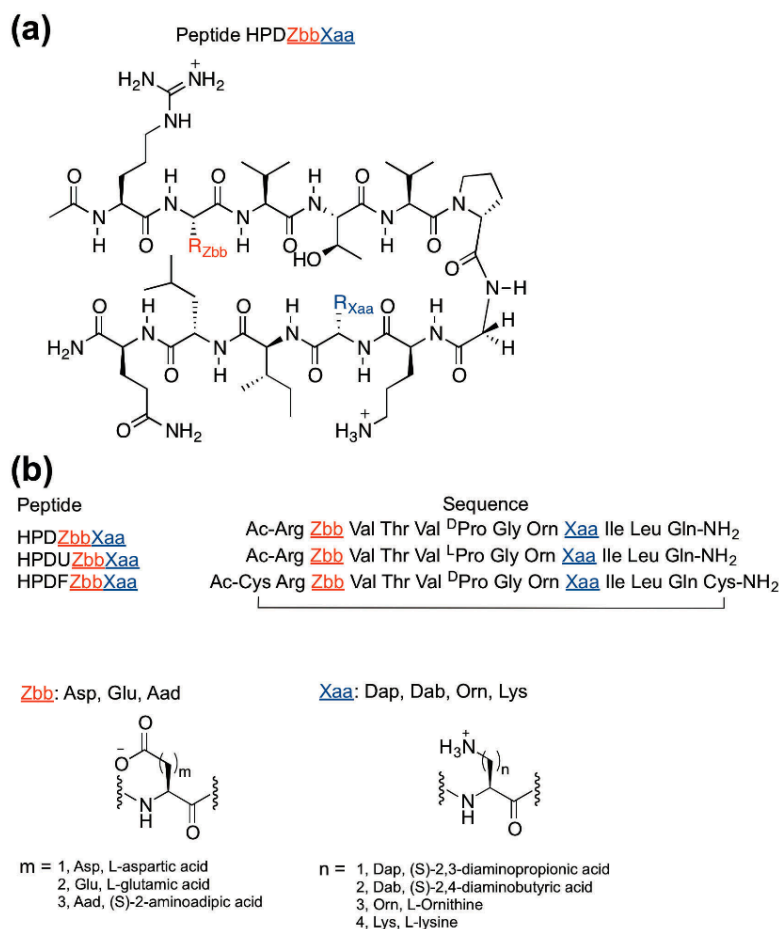


Figure 1. Design of peptides to study the effects of charged amino acid side-chain length on diagonal cross-strand ion-pairing interactions: **(a)** the chemical structure of the experimental HPDZbbXaa peptides; **(b)** the sequences of the experimental HPDZbbXaa peptides, the fully unfolded reference HPDUZbbXaa peptides, and the fully folded reference HPDFZbbXaa peptides.

The peptides were synthesized by solid-phase peptide synthesis using Fmoc-based chemistry [39,40]. The intramolecular disulfide bond in the fully folded reference HPDFZbbXaa peptides was formed via charcoal-mediated air oxidation [41]. All peptides were purified by reverse-phase high-performance liquid chromatography (RP-HPLC) to higher than 95% purity (except for HPDAspDap, HPDAspDab, HPDGluDap, HPDGluDab, and HPDAadDab, which were purified to higher than 90% purity), and confirmed by matrix-assisted laser desorption ionization time-of-flight (MALDI-TOF) mass spectrometry.

2.2. Peptide Structure Characterization by NMR Spectroscopy

The peptides were analyzed by two-dimensional NMR spectroscopy including total correlation spectroscopy (TOCSY), double-quantum filtered-correlated spectroscopy (DQF-COSY), and rotating-frame nuclear Overhauser spectroscopy (ROESY) at 298 K. Sequence-specific assignments of the chemical shifts were performed based on the TOCSY and ROESY spectra (Tables S1–S36) [42]. The number of major spin systems was consistent with the number of residues in each peptide. Weak cross-peaks corresponding to minor conformations were observed in the TOCSY spectra for many of the HPDZbbXaa and HPDUZbbXaa peptides. Since the NMR spectra (chemical shift and peak width) of analogous hairpin peptides did not change with concentration (20 μ M to 10 mM) [3,19,43,44], the peptides in this study (0.9–10.1 mM) should not aggregate in solution. Accordingly, the experimental data should reflect the intramolecular interactions with minimal interference from intermolecular interactions.

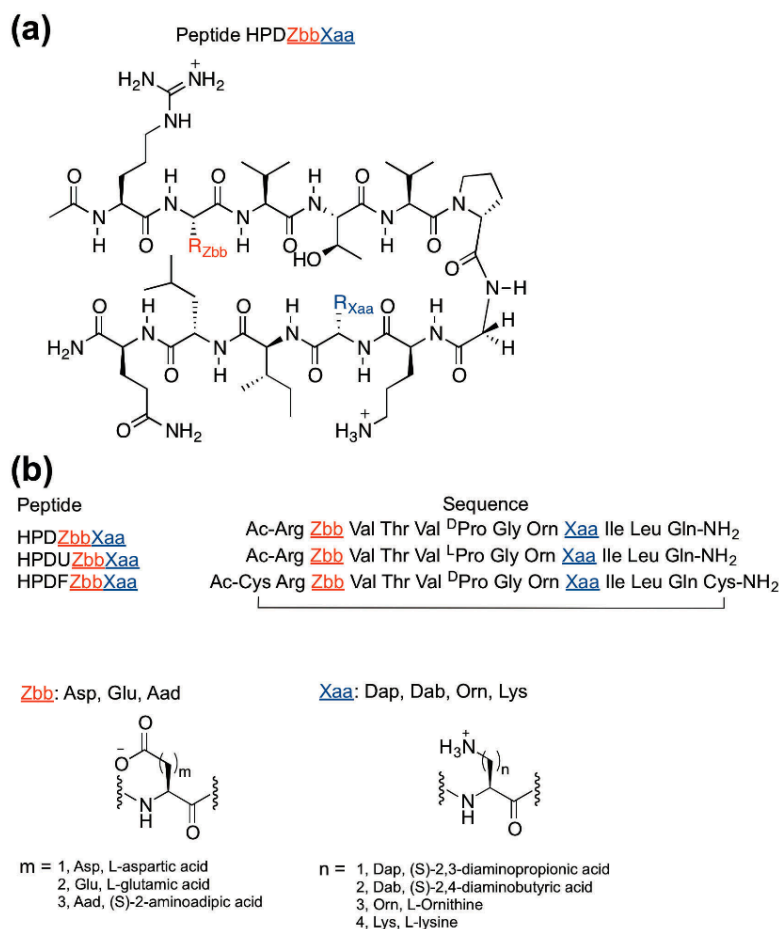


圖 1. 研究帶電胺基酸側鏈長度對對角線交叉鏈離子對交互作用影響的勝肽設計：(a) 實驗 HPDZbbXaa 勝肽的化學結構；(b) 實驗 HPDZbbXaa 勝肽、完全展開的參考 HPDUZbbXaa 勝肽和完全折疊的參考 HPDFZbbXaa 勝肽、完全展開的參考 HPDUZbbXaa 勝肽和完全折疊的參考 HPDFZbbXaa 勝肽的序列。

這些勝肽是使用基於 Fmoc 的化學方法透過固相勝肽合成法合成的 [39,40]。完全摺疊的參考 HPDFZb-bXaa 勝肽中的分子內二硫鍵是透過木炭介導的空氣氧化形成的[41]。所有勝肽均經反相高效液相層析 (RP-HPLC) 純化至高於 95% 的純度 (HPDAspDap、HPDAspDab、HPDGluDap、HPDGluDab 和 HPDAadDab 除外，其純度高於 90%)，並透過基質輔助雷射解電離飛行時間 (MALDI-TOF) 質譜法進行確認飛行時間 (MALDI-TOF)。

2.2. Peptide Structure Characterization by NMR Spectroscopy

透過二維核磁共振波譜分析勝肽，包括全相關波譜 (TOCSY)、雙量子濾波相關波譜 (DQF-COSY) 和 298 K 的旋轉框架核歐沃豪瑟波譜 (ROESY)。根據 TOCSY 和 ROESY 波譜進行化學位移的序列特异性分配 (表 S1-S36) [42]。主要自旋系統的數量與每個勝肽中殘基的數量一致。在許多 HPDZbbXaa 和 HPDUZbbXaa 勝肽的 TOCSY 譜中觀察到與次要構象相對應的弱交叉峰。由於類似髮夾勝肽的 NMR 光譜 (化學位移和峰寬) 不隨濃度 (20 μ M 至 10 mM) 變化[3,19,43,44]，因此本研究中的勝肽 (0.9–10.1 mM) 不應在溶液中聚集。因此，實驗數據應反映分子內相互作用，而分子間相互作用的干擾最小。

The chemical shift dispersion of the peptides was evaluated using the chemical shift ranges of the HN and H α protons, and the standard deviations of the average chemical shifts of the HN and H α protons (Table 1). The broader the chemical shift range, the higher the chemical shift dispersion. Additionally, the higher the standard deviation of the average chemical shift, the higher the chemical shift dispersion. For a given peptide, the chemical shift range was broader for the H α signals than for the HN signals (Table 1). Similarly, the standard deviation of the average chemical shift was higher for the H α signals than for the HN signals. The HN and H α chemical shift ranges followed the trend HPDFZbbXaa > HPDZbbXaa > HPDUZbbXaa, except for the HN chemical shift range for HPDAspDap versus HPDUAspDap (Table 1). Nonetheless, the H α chemical shift range for this exception followed the aforementioned general trend. The standard deviation of the average chemical shifts of the HN and H α protons followed the trend HPDFZbbXaa > HPDZbbXaa > HPDUZbbXaa, except for the standard deviation of the average HN chemical shifts for HPDAspDap versus HPDUAspDap (Table 1). Nevertheless, the standard deviation of the average H α chemical shift for this exception followed the aforementioned general trend. The exception in the HN chemical shift dispersion for HPDAspDap versus HPDUAspDap was most likely because the HN chemical shift is more sensitive to the neighboring residues in the primary sequence, whereas the H α chemical shift is more sensitive to the structural environment [45]. Accordingly, the HN chemical shift dispersion could be either higher than or similar to the H α chemical shift dispersion in the unfolded form, whereas the HN chemical shift dispersion could be lower than the H α chemical shift dispersion in the folded form [45], thereby resulting in the exception in the HN chemical shift dispersion for HPDAspDap versus HPDUAspDap. Overall, the chemical shift dispersion of peptides with a given Zbb2-Xaa9 pair followed the trend HPDFZbbXaa > HPDZbbXaa > HPDUZbbXaa (Tables 1 and S1–S36). Since the higher the chemical shift dispersion, the higher the folded population [45], these results were consistent with the intended designs.

Table 1. The chemical shift range and average chemical shift of the δ HN and δ H α for the HPDFZbbXaa, HPDZbbXaa, and HPDUZbbXaa peptides.

Peptide	δ HN Range ¹ (ppm)	Average δ HN (ppm)	δ H α Range ¹ (ppm)	Average δ H α (ppm)
HPDFAspDap	1.145	8.697 $\pm$ 0.336	1.428	4.664 $\pm$ 0.426
HPDAspDap	0.728	8.406 $\pm$ 0.195	1.009	4.374 $\pm$ 0.276
HPDUAspDap	0.769	8.366 $\pm$ 0.196	0.782	4.351 $\pm$ 0.209
HPDFAspDab	1.169	8.662 $\pm$ 0.339	1.407	4.649 $\pm$ 0.406
HPDAspDab	0.665	8.386 $\pm$ 0.171	0.902	4.368 $\pm$ 0.265
HPDUAspDab	0.570	8.378 $\pm$ 0.153	0.693	4.318 $\pm$ 0.177
HPDFAspOrn	1.124	8.637 $\pm$ 0.333	1.397	4.622 $\pm$ 0.391
HPDAspOrn	0.530	8.346 $\pm$ 0.150	0.895	4.338 $\pm$ 0.250
HPDUAspOrn	0.429	8.317 $\pm$ 0.133	0.723	4.279 $\pm$ 0.193
HPDFAspLys	1.108	8.625 $\pm$ 0.329	1.373	4.608 $\pm$ 0.384
HPDAspLys	0.454	8.326 $\pm$ 0.135	0.878	4.328 $\pm$ 0.244
HPDUAspLys	0.432	8.302 $\pm$ 0.127	0.718	4.276 $\pm$ 0.192
HPDFGluDap	1.119	8.740 $\pm$ 0.330	1.506	4.673 $\pm$ 0.421
HPDGluDap	0.795	8.482 $\pm$ 0.206	1.135	4.404 $\pm$ 0.294
HPDUGluDap	0.574	8.398 $\pm$ 0.172	0.772	4.326 $\pm$ 0.187
HPDFGluDab	1.192	8.706 $\pm$ 0.337	1.467	4.653 $\pm$ 0.404
HPDGluDab	0.626	8.417 $\pm$ 0.179	1.008	4.398 $\pm$ 0.278
HPDUGluDab	0.373	8.368 $\pm$ 0.126	0.514	4.299 $\pm$ 0.148
HPDFGluOrn	1.166	8.695 $\pm$ 0.343	1.457	4.640 $\pm$ 0.394
HPDGluOrn	0.584	8.427 $\pm$ 0.176	0.977	4.369 $\pm$ 0.258
HPDUGluOrn	0.324	8.348 $\pm$ 0.106	0.494	4.289 $\pm$ 0.140
HPDFGluLys	1.217	8.686 $\pm$ 0.360	1.452	4.630 $\pm$ 0.390

使用 HN 和 H α 質子的化學位移範圍以及 HN 和 H α 質子的平均化學位移的標準偏差評估勝肽的化學位移分散度 (表 1)。化學位移範圍越寬，化學位移離散度越高。另外，平均化學位移的標準差越高，化學位移離散度越高。對於給定的勝肽，H α 訊號的化學位移範圍比 HN 訊號更寬 (表 1)。類似地，H α 訊號的平均化學位移的標準差高於 HN 訊號。HN 和 H α 化學位移範圍遵循 HPDFZbbXaa > HPDZbbXaa > HPDUZbbXaa 的趨勢，但 HPDAspDap 與 HPDUAspDap 的 HN 化學位移範圍除外 (表 1)。儘管如此，此異常的 H α 化學位移範圍遵循上述總體趨勢。HN 和 H α 質子的平均化學位移的標準差遵循 HPDFZbbXaa > HPDZbbXaa > HPDUZbbXaa 的趨勢，但 HPDAspDap 與 HPDUAspDap 的平均 HN 化學位移的標準差除外 (表 1)。然而，該例外的平均 H α 化學位移的標準差遵循上述總體趨勢。HPDAspDap 與 HPDUAspDap 的 HN 化學位移離散度的例外很可能是因為 HN 化學位移對一級序列中的相鄰殘基更敏感，而 H α 化學位移對結構環境更敏感 [45]。因此，HN 化學位移色散可能高於或類似於未折疊形式的 H α 化學位移色散，而 HN 化學位移色散可能低於折疊形式的 H α 化學位移色散 [45]，從而導致 HPDAspDap 與 HPDUAspDap 的 HN 化學位移色散存在例外。總體而言，具有給定 Zbb2-Xaa9 對的勝肽的化學位移分散遵循 HPDFZbbXaa > HPDZbbXaa > HPDUZbbXaa 的趨勢 (表 1 和 S1-S36)。由於化學位移色散越高，折疊群體就越高 [45]，這些結果與預期設計一致。

表 1. HPDFZb-bXaa、HPDZbbXaa 和 HPDUZbbXaa 勝肽的 δ HN 和 δ H α 的化學位移範圍和平均化學位移。

Peptide	δ HN Range ¹ (ppm)	Average δ HN (ppm)	δ H α Range ¹ (ppm)	Average δ H α (ppm)
HPDFAspDap	1.145	8.697 $\pm$ 0.336	1.428	4.664 $\pm$ 0.426
HPDAspDap	0.728	8.406 $\pm$ 0.195	1.009	4.374 $\pm$ 0.276
HPDUAspDap	0.769	8.366 $\pm$ 0.196	0.782	4.351 $\pm$ 0.209
HPDFAspDab	1.169	8.662 $\pm$ 0.339	1.407	4.649 $\pm$ 0.406
HPDAspDab	0.665	8.386 $\pm$ 0.171	0.902	4.368 $\pm$ 0.265
HPDUAspDab	0.570	8.378 $\pm$ 0.153	0.693	4.318 $\pm$ 0.177
HPDFAspOrn	1.124	8.637 $\pm$ 0.333	1.397	4.622 $\pm$ 0.391
HPDAspOrn	0.530	8.346 $\pm$ 0.150	0.895	4.338 $\pm$ 0.250
HPDUAspOrn	0.429	8.317 $\pm$ 0.133	0.723	4.279 $\pm$ 0.193
HPDFAspLys	1.108	8.625 $\pm$ 0.329	1.373	4.608 $\pm$ 0.384
HPDAspLys	0.454	8.326 $\pm$ 0.135	0.878	4.328 $\pm$ 0.244
HPDUAspLys	0.432	8.302 $\pm$ 0.127	0.718	4.276 $\pm$ 0.192
HPDFGluDap	1.119	8.740 $\pm$ 0.330	1.506	4.673 $\pm$ 0.421
HPDGluDap	0.795	8.482 $\pm$ 0.206	1.135	4.404 $\pm$ 0.294
HPDUGluDap	0.574	8.398 $\pm$ 0.172	0.772	4.326 $\pm$ 0.187
HPDFGluDab	1.192	8.706 $\pm$ 0.337	1.467	4.653 $\pm$ 0.404
HPDGluDab	0.626	8.417 $\pm$ 0.179	1.008	4.398 $\pm$ 0.278
HPDUGluDab	0.373	8.368 $\pm$ 0.126	0.514	4.299 $\pm$ 0.148
HPDFGluOrn	1.166	8.695 $\pm$ 0.343	1.457	4.640 $\pm$ 0.394
HPDGluOrn	0.584	8.427 $\pm$ 0.176	0.977	4.369 $\pm$ 0.258
HPDUGluOrn	0.324	8.348 $\pm$ 0.106	0.494	4.289 $\pm$ 0.140
HPDFGluLys	1.217	8.686 $\pm$ 0.360	1.452	4.630 $\pm$ 0.390

Table 1. Cont.

Peptide	δHN Range ¹ (ppm)	Average δHN (ppm)	$\delta\text{H}\alpha$ Range ¹ (ppm)	Average $\delta\text{H}\alpha$ (ppm)
HPDGluLys	0.618	8.427 $\pm$ 0.179	0.959	4.395 $\pm$ 0.237
HPDUGluLys	0.355	8.350 $\pm$ 0.102	0.584	4.278 $\pm$ 0.165
HPDFAadDap	1.135	8.738 $\pm$ 0.335	1.536	4.681 $\pm$ 0.424
HPDAadDap	0.793	8.476 $\pm$ 0.211	1.140	4.410 $\pm$ 0.296
HPDUAadDap	0.587	8.298 $\pm$ 0.169	0.782	4.244 $\pm$ 0.190
HPDFAadDab	1.183	8.702 $\pm$ 0.344	1.458	4.654 $\pm$ 0.402
HPDAadDab	0.675	8.445 $\pm$ 0.192	0.958	4.390 $\pm$ 0.268
HPDUAadDab	0.367	8.355 $\pm$ 0.116	0.515	4.300 $\pm$ 0.148
HPDFAadOrn	1.182	8.693 $\pm$ 0.340	1.442	4.638 $\pm$ 0.390
HPDAadOrn	0.535	8.405 $\pm$ 0.164	0.906	4.359 $\pm$ 0.247
HPDUAadOrn	0.273	8.333 $\pm$ 0.092	0.513	4.261 $\pm$ 0.163
HPDFAadLys	1.207	8.687 $\pm$ 0.357	1.455	4.629 $\pm$ 0.387
HPDAadLys	0.599	8.399 $\pm$ 0.172	0.927	4.360 $\pm$ 0.249
HPDUAadLys	0.276	8.317 $\pm$ 0.084	0.530	4.255 $\pm$ 0.165

¹ The chemical shift range was the difference between the chemical shift of the most downfield signal and the chemical shift of the most upfield signal for the proton of interest.

The β -hairpin structure of the peptides was confirmed by the chemical shift deviation of the $\text{H}\alpha$ signals, $^3J_{\text{NH}\alpha}$ spin–spin coupling constants, and cross-strand NOE signals. The $\text{H}\alpha$ chemical shift deviation ($\Delta\delta\text{H}\alpha$) is the difference between the $\text{H}\alpha$ signal of the residue of interest and the corresponding $\text{H}\alpha$ signal of the residue in the random coil conformation. The fully unfolded reference peptides were assumed to be random coil in this study [3,19,25–27,37]. A positive $\Delta\delta\text{H}\alpha$ value represents an extended β -sheet conformation [46,47]. Residues Zbb2 through Val5 and Orn8 through Leu11 showed positive $\Delta\delta\text{H}\alpha$ values for all of the experimental HPDZbbXaa peptides and all the fully folded reference HPDFZbbXaa peptides (Figures 2, S1 and S2), consistent with an extended β -strand conformation for these strand residues. The $\Delta\delta\text{H}\alpha$ values of terminal residues Arg1 and Gln12 for the experimental peptides were close to zero (Figures 2b and S1), most likely resulting from the end fraying effect [3,23,25–27]. The $\Delta\delta\text{H}\alpha$ values of Gly7 were negative or close to zero for all peptides, consistent with a turn conformation for Gly7 [43]. In general, residues in the strand regions (residues 2–5 and 8–11) showed more positive $\Delta\delta\text{H}\alpha$ values for the fully folded reference peptides, compared to those in the corresponding experimental peptides, suggesting that the fully folded reference peptides exhibited a higher β -hairpin population than the corresponding experimental peptides. This was consistent with the intended designs.

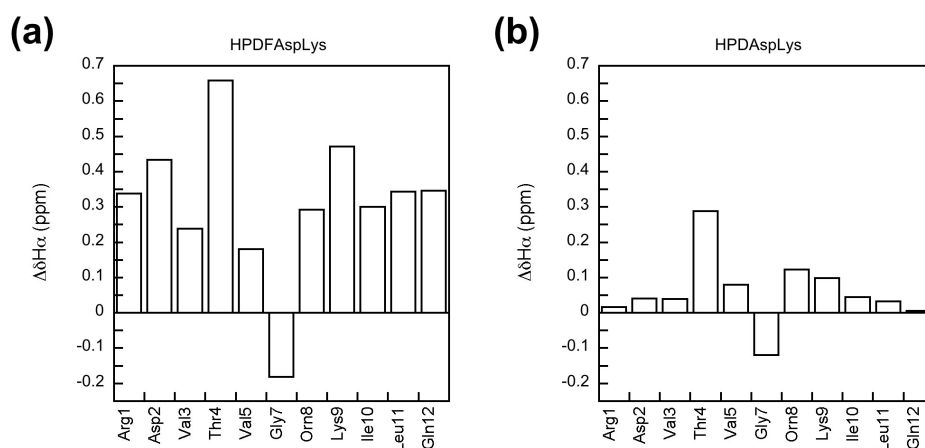


Figure 2. The chemical shift deviation ($\Delta\delta\text{H}\alpha$) for the residues in peptides HPDFAspLys (a) and HPDAspLys (b).

表 1. Cont.

Peptide	$\delta\text{HN Range}^1$ (ppm)	Average δHN (ppm)	$\delta\text{H}\alpha \text{ Range}^1$ (ppm)	Average $\delta\text{H}\alpha$ (ppm)
HPDGluLys	0.618	8.427 ± 0.179	0.959	4.395 ± 0.237
HPDUGluLys	0.355	8.350 ± 0.102	0.584	4.278 ± 0.165
HPDFAadDap	1.135	8.738 ± 0.335	1.536	4.681 ± 0.424
HPDAadDap	0.793	8.476 ± 0.211	1.140	4.410 ± 0.296
HPDUadDap	0.587	8.298 ± 0.169	0.782	4.244 ± 0.190
HPDFAadDab	1.183	8.702 ± 0.344	1.458	4.654 ± 0.402
HPDAadDab	0.675	8.445 ± 0.192	0.958	4.390 ± 0.268
HPDUadDab	0.367	8.355 ± 0.116	0.515	4.300 ± 0.148
HPDFAadOrn	1.182	8.693 ± 0.340	1.442	4.638 ± 0.390
HPDAadOrn	0.535	8.405 ± 0.164	0.906	4.359 ± 0.247
HPDUadOrn	0.273	8.333 ± 0.092	0.513	4.261 ± 0.163
HPDFAadLys	1.207	8.687 ± 0.357	1.455	4.629 ± 0.387
HPDAadLys	0.599	8.399 ± 0.172	0.927	4.360 ± 0.249
HPDUadLys	0.276	8.317 ± 0.084	0.530	4.255 ± 0.165

¹ 化學位移範圍是感興趣質子的最下場訊號的化學位移與最上場訊號的化學位移之間的差異。

勝肽的 β 髮夾結構透過 $\text{H}\alpha$ 訊號的化學位移偏差、 $^3J_{\text{NH}\alpha}$ 自旋-自旋耦合常數和跨鏈NOE訊號得到證實。 $\text{H}\alpha$ 化學位移偏差($\Delta\delta\text{H}\alpha$)是目標殘基的 $\text{H}\alpha$ 訊號與無規捲曲構像中殘基對應的 $\text{H}\alpha$ 訊號之間的差異。在本研究中，完全展開的參考勝肽被假定為隨機捲曲[3,19,25–27,37]。正的 $\Delta\delta\text{H}\alpha$ 值表示擴展的 β -片構象[46,47]。殘基 Zbb2 至 Val5 和 Orn8 至 Leu11 顯示所有實驗 HPDZbbXaa 肽和所有完全折疊的參考 HPDFZbbXaa 肽的正 $\Delta\delta\text{H}\alpha$ 值(圖 2、S1 和 S2)，與這些鏈殘基的延伸 β 鏈構像一致。實驗勝肽的末端殘基 Arg1 和 Gln12 的 $\Delta\delta\text{H}\alpha$ 值接近零(圖 2b 和 S1)，很可能是由於末端磨損效應造成的[3,23,25–27]。所有勝肽的 Gly7 的 $\Delta\delta\text{H}\alpha$ 值均為負值或接近零，與 Gly7 的轉角構像一致[43]。一般來說，與相應的實驗肽中的殘基相比，完全折疊的參考肽的鏈區域中的殘基(殘基 2-5 和 8-11)顯示出更正的 $\Delta\delta\text{H}\alpha$ 值，表明完全折疊的參考肽比相應的實驗肽表現出更高的 β -髮夾群體。這與預期的設計是一致的。

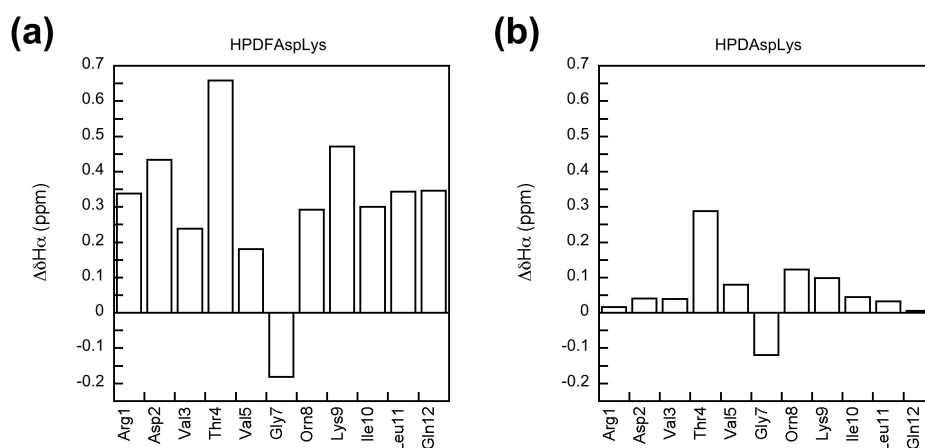


圖 2. 勝肽 HPDAspLys (a) 和 HPDAspLys (b) 中殘基的化學位移偏差 ($\Delta\delta\text{H}\alpha$)。

The $^3J_{NH\alpha}$ spin–spin coupling constants for each residue in the peptides were determined using the DQF-COSY spectra (Tables S37–S45) [48]. The $^3J_{NH\alpha}$ values for the strand residues in the experimental HPDZbbXaa peptides were higher than 7 Hz (Tables S37–S39), suggesting a β -conformation [42,49]. The $^3J_{NH\alpha}$ values for the strand residues in the fully folded reference HPDFZbbXaa peptides were also higher than 7 Hz, and most of the values were higher than those for the corresponding residues in the experimental HPDZbbXaa peptides (Tables 2 and S37–S42). This suggested that the fully folded reference HPDFZbbXaa peptides were more folded than the corresponding experimental HPDZbbXaa peptides. If one disregards the standard deviations, most of the average $^3J_{NH\alpha}$ values followed the trend HPDFZbbXaa > HPDZbbXaa > HPDUZbbXaa (Table 2). Overall, these trends were consistent with the intended designs.

Table 2. The average $^3J_{NH\alpha}$ coupling constant values for the strand residues in the HPDFZbbXaa, HPDZbbXaa, and HPDUZbbXaa peptides ¹.

Zbb	Xaa	HPDFZbbXaa	HPDZbbXaa	HPDUZbbXaa
Asp	Dap	11 ± 0.7	9.8 ± 1.0	10 ± 1.2
Asp	Dab	11 ± 1.6	9.4 ± 1.5	8.3 ± 2.2
Asp	Orn	12 ± 0.7	10 ± 1.0	9.7 ± 0.8
Asp	Lys	12 ± 0.6	10 ± 0.8	10 ± 1.0
Glu	Dap	10 ± 0.9	9.9 ± 0.9	9.6 ± 1.3
Glu	Dab	10 ± 1.1	9.5 ± 2.1	9.2 ± 1.6
Glu	Orn	11 ± 0.6	10 ± 0.6	10 ± 0.6
Glu	Lys	11 ± 1.1	9.6 ± 0.9	9.2 ± 1.0
Aad	Dap	11 ± 0.9	9.7 ± 1.4	9.9 ± 1.0
Aad	Dab	10 ± 1.0	9.4 ± 1.7	9.8 ± 1.0
Aad	Orn	12 ± 0.7	10 ± 0.7	9.3 ± 0.5
Aad	Lys	11 ± 0.5	9.9 ± 0.5	9.6 ± 0.7

¹ The average $^3J_{NH\alpha}$ coupling constant values for the strand residues were calculated without including values for Gly, which is in the β -turn.

All cross-peaks in the ROESY spectra were assigned. Intra-residue, sequential, medium-range, and long-range NOEs (Figures S3–S50) with a number of cross-strand $H\alpha$ -to- $H\alpha$, $H\alpha$ -to-NH, NH-to-NH correlations were observed in the ROESY spectra (Figures S39–S50). Strong sequential $H\alpha_i$ -NH_{i+1} correlations were observed for every strand on all peptides, suggesting an extended β -strand conformation (Figures S39–S50). The cross-strand NOE connectivities obtained from the ROESY spectra further supported the intended β -hairpin conformation for the fully folded reference HPDFZbbXaa peptides and the experimental HPDZbbXaa peptides (Figures S3–S38). For a given Zbb2-Xaa9 combination, the number of cross-strand NOE connectivities (including both side-chain and backbone) followed the trend HPDFZbbXaa > HPDZbbXaa > HPDUZbbXaa (Figures S3–S50). Since more cross-strand NOE connectivities should correspond to a higher fraction β -hairpin structure, these trends are consistent with the intended designs.

2.3. Fraction Folded Population and Folding Free Energy

The fraction folded population and folding free energy (ΔG_{fold}) for each residue on the experimental HPDZbbXaa peptides were derived from the $H\alpha$ chemical shift data (Figures S51 and S52). The ratio of the chemical shift deviation ($\Delta\delta H\alpha$) for a given residue on the experimental HPDZbbXaa peptide to the corresponding value on the fully folded reference HPDFZbbXaa peptide gave the fraction folded population for the residue of interest in the experimental peptide (Figure S51) [3,19,25–27,37]. The average of the fraction folded population for residues at positions 2, 3, 9, and 10 was used to represent the fraction folded population of the corresponding peptide (Table 3) [3,25–27]. These residues were not at the termini of the peptide, at which residues suffered from end fraying effects, nor were these residues adjacent to the turn, at which residues exhibited intrinsically higher fraction folded population [3,19,23,25–27]. Furthermore, both strands, and both hydrogen and

使用 DQF-COSY 譜確定勝肽中每個殘基的 $^3J_{NH\alpha}$ 自旋-自旋耦合常數 (表 S37-S45) [48]。實驗 HPDZbbXaa 勝肽中鏈殘基的 $^3J_{NH\alpha}$ 值高於 7 Hz (表 S37-S39)，顯示存在 β 構象 [42,49]。完全摺疊的參考 HPDFZbbXaa 勝肽中鏈殘基的 $^3J_{NH\alpha}$ 值也高於 7 Hz，且大多數值高於實驗 HPDZbbXaa 勝肽中對應殘基的值 (表 2 和 S37-S42)。這表明完全折疊的參考 HPDFZbbXaa 肽比相應的實驗 HPDZbbXaa 肽折疊得更多。如果忽略標準差，則大多數平均 $^3J_{NH\alpha}$ 值遵循 HPDFZbbXaa > HPDZbbXaa > HPDUZbbXaa 的趨勢 (表 2)。整體而言，這些趨勢與預期設計一致。

表 2. HPDFZbbXaa、HPDZbbXaa 和 HPDUZbbXaa 勝肽¹ 中鏈殘基的平均 $^3J_{NH\alpha}$ 偶聯常數值。

Zbb	Xaa	HPDFZbbXaa	HPDZbbXaa	HPDUZbbXaa
Asp	Dap	11 ± 0.7	9.8 ± 1.0	10 ± 1.2
Asp	Dab	11 ± 1.6	9.4 ± 1.5	8.3 ± 2.2
Asp	Orn	12 ± 0.7	10 ± 1.0	9.7 ± 0.8
Asp	Lys	12 ± 0.6	10 ± 0.8	10 ± 1.0
Glu	Dap	10 ± 0.9	9.9 ± 0.9	9.6 ± 1.3
Glu	Dab	10 ± 1.1	9.5 ± 2.1	9.2 ± 1.6
Glu	Orn	11 ± 0.6	10 ± 0.6	10 ± 0.6
Glu	Lys	11 ± 1.1	9.6 ± 0.9	9.2 ± 1.0
Aad	Dap	11 ± 0.9	9.7 ± 1.4	9.9 ± 1.0
Aad	Dab	10 ± 1.0	9.4 ± 1.7	9.8 ± 1.0
Aad	Orn	12 ± 0.7	10 ± 0.7	9.3 ± 0.5
Aad	Lys	11 ± 0.5	9.9 ± 0.5	9.6 ± 0.7

¹ 計算鏈殘基的平均 $^3J_{NH\alpha}$ 偶聯常數值時不包括位於 β 轉角的 Gly 值。

ROESY 光譜中的所有交叉峰均已指定。在 ROESY 譜中觀察到殘基內、順序、中程和遠程 NOE (圖 S3-S50) 以及許多跨鏈 $H\alpha$ -to- $H\alpha$ 、 $H\alpha$ -to-NH、NH-to-NH 相關性 (圖 S39-S50)。所有勝肽上的每條鏈都觀察到強烈的連續 $H\alpha_i$ -NH_{i+1} 相關性，表明延伸的 β -鏈構象 (圖 S39-S50)。從 ROESY 光譜獲得的跨鏈 NOE 連接進一步支持完全折疊的參考 HPDFZbbXaa 肽和實驗 HPDZbbXaa 肽的預期 β -髮夾構象 (圖 S3-S38)。對於給定的 Zbb2-Xaa 9 組合，跨鏈 NOE 連接數量 (包括側鍊和主鍊) 遵循 HPDFZbbXaa > HPDZbbXaa > HPDUZbbXaa 的趨勢 (圖 S3-S50)。由於更多的跨鏈 NOE 連接應對應於更高分數的 β 髮夾結構，因此這些趨勢與預期設計一致。

2.3. Fraction Folded Population and Folding Free Energy

實驗 HPDZbbXaa 勝肽上每個殘基的摺疊群體分數和摺疊自由能 (ΔG_{fold}) 源自 $H\alpha$ 化學位移資料 (圖 S51 和 S52)。實驗 HPDZbbXaa 勝肽上給定殘基的化學位移偏差 ($\Delta\delta H\alpha$) 與完全折疊參考 HPDFZbbXaa 肽上相應值的比率給出了實驗肽中感興趣殘基的折疊總體分數 (圖 S51) [3,19,25–27,37]。位置 2、3、9 和 10 處殘基的分數折疊群體的平均值用於代表相應勝肽的分數折疊群體 (表 3) [3,25–27]。這些殘基不在勝肽的末端，在該末端，殘基會受到末端磨損效應的影響，這些殘基也不在轉角附近，在轉角處，殘基表現出本質上更高的折疊群體分數 [3,19,23,25–27]。此外，兩條鏈，以及氫和

non-hydrogen bonded positions, were equally represented in the chosen residues [3,25–27]. Similarly, the average of the ΔG_{fold} for residues at positions 2, 3, 9, and 10 was used to represent the ΔG_{fold} of the corresponding peptide (Table 4) [3,25–27]. Since the ΔG_{fold} exhibited the same trends as the fraction folded population (i.e., the higher the fraction folded population, the less positive the ΔG_{fold}), only the fraction folded population data are discussed in detail.

Table 3. The fraction folded population (%) for HPDZbbXaa peptides ¹.

Xaa9	Zbb2		
	Asp	Glu	Aad
Dap	19 ± 4	38 ± 4	47 ± 4
Dab	25 ± 4	42 ± 3	40 ± 3
Orn	17 ± 4	36 ± 3	33 ± 3
Lys	15 ± 5	36 ± 4	36 ± 3

¹ Average value for residues 2, 3, 9, and 10.

Table 4. The folding free energy (ΔG_{fold} , kcal/mol) for HPDZbbXaa peptides ¹.

Xaa9	Zbb2		
	Asp	Glu	Aad
Dap	0.864 ± 0.177	0.303 ± 0.091	0.062 ± 0.085
Dab	0.665 ± 0.143	0.185 ± 0.065	0.231 ± 0.070
Orn	0.975 ± 0.188	0.347 ± 0.075	0.424 ± 0.077
Lys	1.031 ± 0.231	0.333 ± 0.107	0.345 ± 0.086

¹ Average value for residues 2, 3, 9, and 10.

The fraction folded population for HPDZbbXaa peptides was between 15% and 47% with standard deviations within 5% (Table 3). HPDAadDap exhibited the highest fraction folded population among all HPDZbbXaa peptides. HPDAspLys exhibited the least fraction folded population among the peptides. If one disregards HPDAspDap, the fraction folded population of HPDZbbXaa peptides with a given negatively charged residue Zbb2 generally decreased as the side-chain length of the positively charged residue Xaa9 increased. Furthermore, the fraction folded population of HPDZbbXaa peptides with a given positively charged residue Xaa9 generally increased as the side-chain length of the negatively charged residues Zbb2 increased from Asp to Glu but mostly remained similar upon increasing Glu to Aad.

2.4. Diagonal Cross-Strand Zbb-Xaa Interactions

The interaction free energy (ΔG_{int}) for each potential diagonal Zbb2-Xaa9 interaction was derived by double mutant cycle analysis (Table 5) [50]. The double mutant cycle was used to remove the inherent effect of individually incorporating the charged residues Zbb2 and Xaa9 from the effect of simultaneously incorporating both Zbb2 and Xaa9 [25–27]. The energy difference between the ΔG_{fold} of HPDZbbXaa and HPDAlaAla represented the effect of simultaneously incorporating both Zbb2 and Xaa9. The energy difference between the ΔG_{fold} of HPDZbbAla (and HPDAlaXaa) and HPDAlaAla represented the effect of individually incorporating Zbb2 (and Xaa9) on β -hairpin stability. The Zbb2-Xaa9 interaction energy (ΔG_{int}) was derived from the folding free energy for the peptides HPDZbbXaa, HPDZbbAla [38], HPDAlaXaa [38], and HPDAlaAla (Table 5).

非氫鍵位置，在所選殘基中同樣具有代表性[3,25–27]。類似地，位置 2、3、9 和 10 處殘基的 ΔG_{fold} 平均值用於表示對應勝肽的 ΔG_{fold} （表 4）[3,25–27]。由於 ΔG_{fold} 表現出與分數折疊總體相同的趨勢（即，分數折疊總體越高， ΔG_{fold} 的正值越低），因此僅詳細討論分數折疊總體數據。

表 3. HPDZbbXaa 勝肽¹ 的摺疊群體分數 (%)。

Xaa9	Zbb2		
	Asp	Glu	Aad
Dap	19 ± 4	38 ± 4	47 ± 4
Dab	25 ± 4	42 ± 3	40 ± 3
Orn	17 ± 4	36 ± 3	33 ± 3
Lys	15 ± 5	36 ± 4	36 ± 3

¹ 殘基 2、3、9 和 10 的平均值。

表 4. HPDZbbXaa 勝肽¹ 的摺疊自由能 (ΔG_{fold} , kcal/mol)。

Xaa9	Zbb2		
	Asp	Glu	Aad
Dap	0.864 ± 0.177	0.303 ± 0.091	0.062 ± 0.085
Dab	0.665 ± 0.143	0.185 ± 0.065	0.231 ± 0.070
Orn	0.975 ± 0.188	0.347 ± 0.075	0.424 ± 0.077
Lys	1.031 ± 0.231	0.333 ± 0.107	0.345 ± 0.086

¹ 殘基 2、3、9 和 10 的平均值。

HPDZbbXaa 勝肽的摺疊群體分數在 15% 到 47% 之間，標準差在 5% 以內（表 3）。HPDAadDap 在所有 HPDZbbXaa 勝肽中表現出最高的分數折疊群。HPDAspLys 在勝肽中表現出最少分數的折疊群體。如果忽略 HPDAspDap，則具有給定帶負電殘基 Zbb2 的 HPDZbbXaa 勝肽的分數折疊群體通常會隨著帶正電殘基 Xaa9 的側鏈長度增加而減少。此外，具有給定帶正電殘基 Xaa9 的 HPDZbbXaa 勝肽的折疊群體分數通常隨著帶負電殘基 Zbb2 的側鏈長度從 Asp 增加到 Glu 而增加，但在將 Glu 增加到 Aad 時大多保持相似。

2.4. Diagonal Cross-Strand Zbb-Xaa Interactions

每個潛在對角 Zbb2-Xaa9 交互作用的交互作用自由能 (ΔG_{int}) 是透過雙突變體循環分析得出（表 5）[50]。雙突變體循環用於消除單獨摺入帶電殘基 Zbb2 和 Xaa9 的固有影響以及同時摺入 Zbb2 和 Xaa9 的影響 [25–27]。HPDZbbXaa 和 HPDAlaAla 的 ΔG_{fold} 之間的能量差代表同時摺入 Zbb2 和 Xaa9 的效果。HPDZbbAla（和 HPDAlaXaa）和 HPDAlaAla 的 ΔG_{fold} 之間的能量差代表單獨摺入 Zbb2（和 Xaa9）對 β -髮夾穩定性的影響。Zbb2-Xaa9 交互作用能 (ΔG_{int}) 源自於勝肽 HPDZbbXaa、HPDZbbAla [38]、HPDAlaXaa [38] 和 HPDAlaAla 的摺疊自由能（表 5）。

Table 5. The diagonal Zbb2-Xaa9 ion-pairing interaction energy (ΔG_{int} , kcal/mol).

Xaa9	Zbb2		
	Asp	Glu	Aad
Dap	-0.434 ± 0.204	-0.412 ± 0.043	-0.425 ± 0.077
Dab	-0.794 ± 0.139	-0.585 ± 0.058	-0.284 ± 0.030
Orn	-0.375 ± 0.081	-0.291 ± 0.048	-0.023 ± 0.040
Lys	-0.206 ± 0.018	-0.258 ± 0.026	-0.020 ± 0.045

Most of the cross-strand diagonal Zbb2-Xaa9 interactions were apparently stabilizing (Table 5). However, there was no apparent Aad2-Orn9 or Aad2-Lys9 interaction. Nonetheless, the Asp2-Dab9 interaction was the most stabilizing, providing up to -0.794 kcal/mol stabilization. The Zbb2-Dap9 interactions were similar in energy regardless of the negatively charged Zbb2 residue. For the HPDZbbDab and HPDZbbOrn peptides, the ΔG_{int} for the Zbb2-Dab9 and Zbb2-Orn9 interactions followed the trend Asp < Glu < Aad. For the HPDZbbLys peptides, the ΔG_{int} for the Zbb2-Lys9 interaction followed the trend Asp > Glu < Aad. If one disregards the Asp2-Dap9 and Glu2-Dap9 interactions, the ΔG_{int} for the Zbb2-Xaa9 interaction was less stabilizing upon increasing the Xaa9 side-chain length for a given negatively charged Zbb2 residue.

3. Discussion

The effects of side-chain length on diagonal ion-pairing interactions between carboxylate- and ammonium-containing residues in a β -hairpin was investigated. The fraction folded population for the experimental HPDZbbXaa peptides was 15–47% (Table 3). The variation in the fraction folded population of HPDZbbXaa peptides can be rationalized by the intrinsic sheet propensity of the carboxylate- and ammonium-containing amino acids with different side-chain lengths at positions 2 and 9, respectively, and the potential cross-strand diagonal Zbb2-Xaa9 ion-pairing interaction. In general, the fraction folded population for the HPDZbbXaa peptides with a given positively charged residue Xaa9 increased as the side-chain length of the negatively charged residues Zbb2 increased from Asp to Glu but mostly remained similar upon increasing Glu to Aad (Table 3). This trend is similar to the fraction folded population trend for the HPDZbbAla peptides (with the negatively charged residue Zbb at position 2 and Ala at position 9) [38], suggesting that there may be small variations in the diagonal Zbb2-Xaa9 interactions in the HPDZbbXaa peptides. However, the fraction folded population for the HPDZbbXaa peptides with a given negatively charged residue Zbb2 generally decreased as the side-chain length of the positively charged residue Xaa9 increased (Table 3). This trend is opposite of the fraction folded population trend for the HPDAAlaXaa peptides (with Ala at position 2 and the positively charged ammonium-containing residue Xaa at position 9) [38], suggesting variation in the diagonal Zbb2-Xaa9 interaction in the HPDZbbXaa peptides. Indeed, the diagonal Zbb2-Xaa9 interaction varied with side-chain length combination (Table 5).

How do the HPDZbbXaa peptides with a diagonal Zbb2-Xaa9 interaction measure up to the HPTZbbXaa peptides with a lateral Zbb4-Xaa9 interaction [25]? The constituting amino acids for the two sets of peptides were the same, but Thr and Zbb were incorporated at positions 2 and 4, respectively, in the HPTZbbXaa peptides [25]. The fraction folded population of the HPDZbbXaa peptides was mostly lower than that of the corresponding HPTZbbXaa peptides [25], except for HPDGluDab, HPDAadDap, and HPDAadDab. Furthermore, the range for the fraction folded population of the HPDZbbXaa peptides was narrower than that of the HPTZbbXaa peptides (24–63%) [25]. The HPDAadDap ($47 \pm 4\%$) and HPDGluDab ($42 \pm 3\%$) peptides exhibited the highest fraction folded populations for the HPDZbbXaa peptides in this study. Surprisingly, HPTAadDap ($25 \pm 1\%$) and HPTGluDab ($24 \pm 2\%$) exhibited the lowest fraction folded populations for the HPTZbbXaa peptides [25]. These differences may be due to the effect of incorporating the negatively charged residue at position 2 versus 4, the effect of incorporating Thr at

表 5. 對角 Zbb2-Xaa9 離子對交互作用能 (ΔG_{int} , kcal/mol)。

Xaa9	Zbb2		
	Asp	Glu	Aad
Dap	-0.434 ± 0.204	-0.412 ± 0.043	-0.425 ± 0.077
Dab	-0.794 ± 0.139	-0.585 ± 0.058	-0.284 ± 0.030
Orn	-0.375 ± 0.081	-0.291 ± 0.048	-0.023 ± 0.040
Lys	-0.206 ± 0.018	-0.258 ± 0.026	-0.020 ± 0.045

大多數跨鏈對角 Zbb2-Xaa9 交互作用明顯穩定 (表 5)。然而，沒有明顯的 Aad2-Orn 9 或 Aad2-Lys9 相互作用。儘管如此，Asp2-Dab9 相互作用是最穩定的，提供高達 -0.794 kcal/mol 的穩定性。無論 Zbb2 殘基帶負電荷為何，Zbb2-Dap9 相互作用的能量相似。對於 HPDZbbDab 和 HPDZbbOrn 勝肽，Zbb2-Dab9 和 Zbb2-Orn9 相互作用的 ΔG_{int} 遵循 Asp < Glu < Aad 趨勢。對於 HPDZbbLys 勝肽，Zbb2-Lys9 相互作用的 ΔG_{int} 遵循 Asp > Glu < Aad 趨勢。如果忽略 Asp2-Dap9 和 Glu2-Dap9 相互作用，則在增加給定帶負電荷的 Zbb2 殘基的 Xaa9 側鏈長度後，Zbb2-Xaa9 相互作用的 ΔG_{int} 穩定性較差。

三、討論

研究了側鏈長度對 β -髮夾中羧酸根和含鉸殘基之間對角離子對相互作用的影響。實驗 HPDZbbXaa 勝肽的摺疊群體分數為 15-47% (表 3)。HPDZbbXaa 勝肽折疊群體分數的變化可以透過分別在位置 2 和 9 處具有不同側鏈長度的含羧酸鹽和含鉸氨基酸的內在片傾向以及潛在的跨鏈對角 Zbb2-Xaa9 離子對相互作用來合理化。一般而言，隨著帶負電殘基 Zbb2 的側鏈長度從 Asp 增加到 Glu，具有給定帶正電殘基 Xaa9 的 HPDZbbXaa 勝肽的折疊群體分數增加，但在將 Glu 增加到 Aad 後，大部分保持相似 (表 3)。這種趨勢與 HPDZbbAla 肽的分數折疊群體趨勢類似 (帶負電荷的殘基 Zbb 在位置 2，Ala 在位置 9) [38]，表明 HPDZbbXaa 肽中的對角線 Zbb2-Xaa9 相互作用可能存在微小變化。然而，具有給定帶負電殘基 Zbb2 的 HPDZbbXaa 勝肽的折疊群體分數通常隨著帶正電殘基 Xaa9 的側鏈長度增加而減少 (表 3)。此趨勢與 HPDAlaXaa 勝肽的分數摺疊群體趨勢相反 (位置 2 為 Ala，位置 9 為帶正電的含鉸殘基 Xaa) [38]，顯示 HPDZbbXaa 勝肽中對角線 Zbb2-Xaa9 交互作用有變化。事實上，對角 Zbb2-Xaa9 交互作用隨側鏈長度組合而變化 (表 5)。

具有對角線 Zbb2-Xaa9 相互作用的 HPDZbbXaa 肽如何與具有橫向 Zbb4-Xaa9 相互作用的 HPTZbbXaa 肽相媲美 [25]？兩組勝肽的組成胺基酸相同，但 Thr 和 Zbb 分別摻入 HPTZbbXaa 勝肽的第 2 和第 4 位 [25]。除 HPDGluDab、HPDAadDap 和 HPDAadDab 外，HPDZbbXaa 勝肽的折疊群體分數大多低於相應的 HPTZbbXaa 勝肽 [25]。此外，HPDZbbXaa 勝肽的摺疊群體分數範圍比 HPTZbbXaa 勝肽窄 (24-63%) [25]。在本研究中，HPDAadDap ($47 \pm 4\%$) 和 HPDGluDab ($42 \pm 3\%$) 勝肽在 HPDZbbXaa 勝肽中表現出最高的折疊群體分數。令人驚訝的是，HPTAadDap ($25 \pm 1\%$) 和 HPTGluDab ($24 \pm 2\%$) 表現出 HPTZbbXaa 勝肽折疊群體分數最低 [25]。這些差異可能是由於在位置 2 與位置 4 處摻入帶負電殘基的影響，以及在位置 4 處摻入 Thr 的影響。

position 4 versus 2, and the difference in the diagonal Zbb2-Xaa9 interaction and the lateral Zbb4-Xaa9 interaction. The fraction folded populations of HPDAspAla ($8 \pm 5\%$) and HPDGluAla ($19 \pm 5\%$) were lower than those of HPTAspAla ($37 \pm 1\%$) and HPTGluAla ($29 \pm 2\%$) [3,38], respectively, whereas the fraction folded populations of HPDAadAla ($26 \pm 4\%$) and HPTAadAla ($25 \pm 2\%$) were essentially the same [3,38]. Furthermore, the fraction folded populations for HPDAlaDab ($21 \pm 3\%$) and HPDAlaLys ($27 \pm 3\%$) were higher than those for HPTAlaDab ($14 \pm 3\%$) and HPTAlaLys ($22 \pm 3\%$) [3,38], respectively, whereas the fraction folded populations of HPDAlaDap ($19 \pm 6\%$) and HPDAlaOrn ($24 \pm 3\%$) were similar to those of HPTAlaDap ($16 \pm 1\%$) and HPTAlaOrn ($21 \pm 3\%$) [3,38], respectively. Both HPDAadDap and HPDGluDab exhibited relatively high fraction folded populations, most likely because the relatively longer Aad and Glu side chains favor strand formation [38], coupled with reasonably stabilizing diagonal Aad2-Dap9 and Glu2-Dab9 ion-pairing interaction (Table 5), respectively.

The diagonal ion-pairing interaction between carboxylate- and ammonium-containing residues with varying side-chain lengths provided up to 0.79 kcal/mol stabilization in β -hairpins (Table 5). This is on par with the diagonal hydrophobics/ π -related interactions, which provided up to 1.0 kcal/mol stabilization in β -hairpins [24,29–35]. Furthermore, the diagonal ion-pairing interaction in this study also matched the analogous lateral ion-pairing interaction, which provided up to 0.83 kcal/mol stabilization in β -hairpins [25]. Focusing on the encoded amino acids, the diagonal Asp2-Lys9 (-0.21 kcal/mol) and Glu2-Lys9 (-0.26 kcal/mol) interactions were more stabilizing than the lateral Asp4-Lys9 (-0.11 kcal/mol), Glu4-Lys9 (-0.02 kcal/mol), Lys4-Asp9 (destabilizing), and Lys4-Glu9 (destabilizing) interactions [3,27]. The diagonal Asp2-Dab9 and Glu2-Dab9 interactions were the most stabilizing among the diagonal Zbb2-Xaa9 pairs. However, the lateral Asp4-Dab9 and Glu4-Dab9 pairs showed no apparent interaction in our previous study [25]. This may be due to the difference in the relative positioning of the potentially interacting residues for the diagonal and lateral interactions. Based on the solution structure of an analogue of the parent YKL peptide [51], the C α -C α distance for the diagonal Zbb2-Xaa9 pair is 5.91 Å (591 pm), whereas the C α -C α distance for the lateral Zbb4-Xaa9 pair is 3.96 Å (396 pm). Furthermore, the C β -C β distance for the diagonal Zbb2-Xaa9 pair is 4.85 Å (485 pm), whereas the C β -C β distance for the lateral Zbb4-Xaa9 pair is 5.41 Å (541 pm). According to this solution structure [51], diagonal pairs point toward each other, whereas lateral pairs point away from each other (Figure 3).

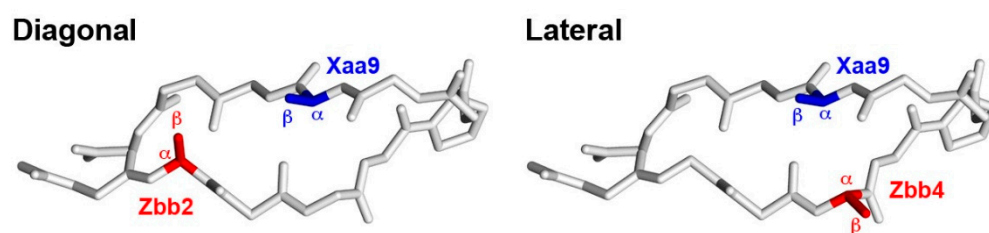


Figure 3. The backbone of the solution structure of an analogue of the parent YKL peptide (pdb code 1JY9 [51]) along with the intended diagonal (**left**) and lateral (**right**) interacting residues. The backbone and DPro side chains are shown in white. The C α and C β of the negatively charged residues are shown in red, and C α and C β of the positively charged residues are shown in blue.

The ΔG_{int} for the diagonal Zbb2-Dap9 interactions were similar regardless of the negatively charged Zbb2 side-chain length. The Asp2-Dap9 ion pair should suffer the least side-chain conformational entropic penalty for ion pair formation, compared to the other Zbb2-Dap9 ion-pairs, because the Asp side-chain length is the shortest among the carboxylate-containing residues. Increasing the carboxylate-containing Zbb2 side-chain length would increase the side-chain conformational entropic penalty for the Zbb2-Dap9 ion-pair formation but would also increase the number of side-chain conformations that could accommodate a Zbb2-Dap9 ion-pair to compensate for the increased entropic penalty for ion-pair formation, resulting in similar ΔG_{int} values. Increasing the Dap side-chain

位置 4 與 2，以及對角 Zbb2-Xaa9 交互作用和橫向 Zbb4-Xaa9 交互作用的差異。HPDAspAla ($8 \pm 5\%$) 和 HPDGluAla ($19 \pm 5\%$) 的折疊群體分數分別低於 HPTAspAla ($37 \pm 1\%$) 和 HPTGluAla ($29 \pm 2\%$) [3,38]，而 HPDADAadla (2665) [3,38] 2%) 本質上是相同的 [3,38]。此外，HPDAlaDab ($21 \pm 3\%$) 和 HPDAlaLys ($27 \pm 3\%$) 的摺疊群體分數分別高於 HPTAlaDab ($14 \pm 3\%$) 和 HPTAlaLys ($22 \pm 3\%$) [3,38]，而 HPDAlaD (1969%) 的折疊群體分數 (NDA8]，而 HPDAla ($19969\% \pm 3\%$) 分別與 HPTAlaDap ($16 \pm 1\%$) 和 HPTAlaOrn ($21 \pm 3\%$) 相似 [3,38]。HPDAadDap 和 HPDGluDab 均表現出相對較高的折疊群體比例，很可能是因為相對較長的 Aad 和 Glu 側鏈有利於鏈形成 [38]，再加上合理穩定的對角線 Aad2-Dap9 和 Glu2-Dab9 離子對相互作用 (表 5)。

具有不同側鏈長度的含羧酸根和含銨殘基之間的對角離子對相互作用在 β -髮夾中提供高達 0.79 kcal/mol 的穩定性 (表 5)。這與對角疏水/ π 相關的相互作用相當，它在 β -髮夾中提供高達 1.0 kcal/mol 的穩定性 [24,29–35]。此外，本研究中的對角離子對相互作用也與類似的橫向離子對相互作用相匹配，後者在 β -髮夾中提供高達 0.83 kcal/mol 的穩定性 [25]。著眼於編碼的胺基酸，對角 Asp2-Lys9 (-0.21 kcal/mol) 和 Glu2-Lys9 (-0.26 kcal/mol) 交互作用比橫向 Asp4-Lys9 (-0.11 kcal/mol)、Glu4-Lys9 ($\{v4\text{-Lys9}\} \{v22\text{Lys4-Asp9}\}$ (不穩定) 和 Lys4-Glu9 (不穩定) 交互作用 [3,27]。對角的 Asp2-Dab9 和 Glu2-Dab9 交互作用是對角的 Zbb2-Xaa9 對中最穩定的。然而，在我們先前的研究中，側向 Asp4-Dab9 和 Glu4-Dab9 對並沒有顯示出明顯的交互作用 [25]。這可能是由於對角相互作用和橫向相互作用的潛在相互作用殘基的相對定位的差異。根據母體 YKL 勝肽類似物的溶液結構 [51]，對角 Zbb2-Xaa9 對的 $C\alpha$ - $C\alpha$ 距離為 $5.91 \text{ \AA}$ (591 pm)，而橫向 Zbb4-Xaa9 對的 $C\alpha$ - $C\alpha$ 距離為 3.9696 。此外，對角 Zbb2-Xaa9 對的 $C\beta$ - $C\beta$ 距離為 $4.85 \text{ \AA}$ (485 pm)，而橫向 Zbb4-Xaa9 對的 $C\beta$ - $C\beta$ 距離為 $5.41 \text{ \AA}$ (541 pm)。根據此解結構 [51]，對角線對彼此指向，而橫向對彼此遠離 (圖 3)。

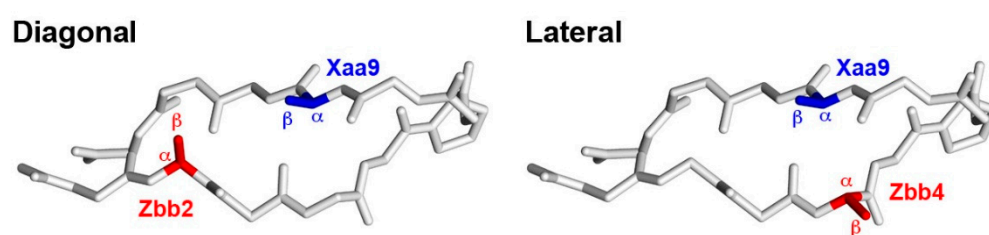


圖 3. 母體 YKL 勝肽類似物 (pdb 代碼 1JY9 [51]) 的溶液結構骨架以及預期的對角線 (左) 和橫向 (右) 相互作用殘基。主鏈和 DPro 側鏈以白色顯示。帶負電的殘基的 $C\alpha$ 和 $C\beta$ 以紅色顯示，帶正電的殘基的 $C\alpha$ 和 $C\beta$ 以藍色顯示。

無論帶負電荷的 Zbb2 側鏈長度為何，對角 Zbb2-Dap9 交互作用的 ΔG_{int} 都是相似的。與其他 Zbb2-Dap9 離子對相比，Asp2-Dap9 離子對在離子對形成過程中受到的側鏈構象熵損失應最小，因為 Asp 側鏈長度在含羧酸根的殘基中最短。增加含羧酸根的 Zbb2 側鏈長度會增加 Zbb2-Dap9 離子對形成的側鏈構象熵損失，但也會增加可容納 Zbb2-Dap9 離子對的側鏈構象數量，以補償離子對形成增加的熵損失，從而產生類似的 ΔG_{int} 值。增加 Dap 側鏈

length in the Asp2-Dap9 ion-pair to the Asp2-Dab9 ion pair resulted in the most stabilizing diagonal ion-pairing interaction, suggesting that the increase in the number of side-chain conformations that could accommodate an Asp2-Dab9 ion-pair outweighed the increase in the side-chain conformational entropic penalty for ion-pair formation upon lengthening Dap9 to Dab9. Further side-chain lengthening of either Asp2 or Dab9 (or both) resulted in a drop in stabilization for the Zbb2-Xaa9 interaction, suggesting that the increase in the number of conformations that could accommodate a Zbb2-Xaa9 ion-pair could no longer compensate for the increase in the entropic penalty for ion-pair formation. As such, the diagonal Asp2-Dab9 ion-pairing interaction most likely represents an optimal balance between side-chain conformational entropic penalty and the number of side-chain conformations to support the Asp2-Dab9 interaction. This is in sharp contrast to the length matching requirements for the lateral Zbb4-Xaa9 interaction [25], in which the longer side chains most likely interact through hydrophobics and the shorter side chains most likely interact through electrostatics.

4. Materials and Methods

4.1. Peptide Synthesis and Purification

Peptides were synthesized by solid-phase peptide synthesis using Fmoc-based chemistry [39,40]. The disulfide bond in the HPDFZbbXaa peptides was formed via charcoal-mediated air oxidation [41]. All peptides were purified by reverse-phase high-performance liquid chromatography (RP-HPLC) to higher than 95% purity (except for HPDAspDap, HPDAspDab, HPDGluDap, HPDGluDab, and HPDAadDab, which were purified to higher than 90% purity). The identity of the peptides was confirmed by matrix-assisted laser desorption ionization time-of-flight mass spectroscopy (MALDI-TOF). More detailed procedures and peptide characterization data are provided in the Supplementary Materials.

4.2. Chemical Shift Deviation

Purified peptides were dissolved in a H₂O/D₂O (9:1 ratio by volume) in the presence of 50 mM sodium deuterioacetate buffer (pH 5.5, uncorrected) to give peptide concentrations of 0.9–10.1 mM. 2-Dimethyl-2-silapentane-5-sulfonate was added to the sample as an internal reference. All NMR experiments were performed on a Bruker AV III 800 MHz spectrometer. Phase-sensitive total correlation spectroscopy (TOCSY) [52], rotating-frame nuclear Overhauser effect spectroscopy (ROESY) [53], and double-quantum filtered-correlated spectroscopy (DQF-COSY) [54] experiments were performed by collecting 2048 points in f2 with 4–8 scans and 256–512 points in f1 at 298 K. Solvent suppression was achieved by the WATERGATE solvent suppression sequence [55]. TOCSY and ROESY experiments employed a spin locking field of 10 kHz. Mixing times of 60 and 200 ms were used for the TOCSY and ROESY experiments, respectively.

4.3. Nuclear Magnetic Resonance Spectroscopy

Sequence-specific assignments for all peptides were completed by using the 2D-NMR spectra (TOCSY and ROESY). The chemical shift deviation ($\Delta\delta H\alpha$) for each residue of the experimental peptide ($\Delta\delta H\alpha(\text{exp})$) and the folded reference peptide ($\Delta\delta H\alpha(\text{F})$) was derived using Equations (1) and (2), respectively [46]. $\delta H\alpha(\text{exp})$ is the chemical shift for the residue of interest on the experimental peptide. $\delta H\alpha(\text{U})$ is the chemical shift for the corresponding residue of interest on the fully unfolded reference peptide. $\delta H\alpha(\text{F})$ is the chemical shift for the corresponding residue of interest on the fully folded reference peptide.

$$\Delta\delta H\alpha(\text{exp}) = \delta H\alpha(\text{exp}) - \delta H\alpha(\text{U}) \quad (1)$$

$$\Delta\delta H\alpha(\text{F}) = \delta H\alpha(\text{F}) - \delta H\alpha(\text{U}) \quad (2)$$

Asp2-Dap9 離子對到 Asp2-Dab9 離子對的長度導致最穩定的對角離子對相互作用，表明可容納 Asp2-Dab9 離子對的側鏈構象數量的增加超過了將 Dap9 延長至 Dab9 時離子對形成的側鏈構象熵損失的增加。Asp2 或 Dab9（或兩者）的進一步側鏈延長導致 Zbb2-Xaa9 相互作用的穩定性下降，這表明可容納 Zbb2-Xaa9 離子對的構象數量的增加不再能夠補償離子對形成的熵罰的增加。因此，對角 Asp2-Dab9 離子對相互作用很可能代表側鏈構象熵罰分和支持 Asp2-Dab9 相互作用的側鏈構象數量之間的最佳平衡。這與橫向 Zbb4-Xaa9 相互作用的長度匹配要求形成鮮明對比 [25]，其中較長的側鏈最有可能透過疏水性相互作用，而較短的側鏈最有可能透過靜電相互作用。

4. 材料與方法

4.1. Peptide Synthesis and Purification

使用基於 Fmoc 的化學方法透過固相勝肽合成法合成勝肽[39,40]。HPDFZbbXaa 勝肽中的二硫鍵是透過木炭介導的空氣氧化形成的[41]。所有勝肽均經反相高效液相層析 (RP-HPLC) 純化至純度高於 95% (HPDAspDap、HPDAspDab、HPDGluDap、HPDGluDab 和 HPDAadDab 除外，其純度高於 90%)。透過基質輔助雷射解吸電離飛行時間質譜 (MALDI-TOF) 確認了勝肽的身份。補充資料中提供了更詳細的程序和勝肽表徵數據。

4.2. Chemical Shift Deviation

在 50 mM 氘代乙酸鈉緩衝液 (pH 5.5, 未校正) 存在下，將純化的勝肽溶解在 H₂O/D₂O (體積比 9:1) 中，得到 0.9–10.1 mM 的勝肽濃度。將 2-二甲基-2-矽戊烷-5-磺酸酯加入樣品中作為內參。所有 NMR 實驗均在 Brüker AV III 800 MHz 光譜儀上進行。相敏全相關光譜 (TOCSY) [52]、旋轉框架核奧豪塞效應光譜 (ROESY) [53] 和雙量子濾波相關光譜 (DQF-COSY) [54] 實驗是透過在 298 K 下透過 4–8 次掃描收集 f₂ 中的 2048 個點和 f₁ 中的 25 個點來進行的 25 個點。透過 WATERGATE 溶劑抑制序列實現了溶劑抑制[55]。TOCSY 和 ROESY 實驗採用 10 kHz 的自旋鎖定場。TOCSY 和 ROESY 實驗分別使用 60 毫秒和 200 毫秒的混合時間。

4.3. Nuclear Magnetic Resonance Spectroscopy

所有勝肽的序列特异性分配均透過使用 2D-NMR 譜 (TOCSY 和 ROESY) 完成。實驗勝肽 ($\Delta\delta H\alpha(\text{exp})$) 和折疊參考勝肽 ($\Delta\delta H\alpha(\text{F})$) 的每個殘基的化學位移偏差 ($\Delta\delta H\alpha$) 分別使用方程式 (1) 和 (2) 來得出 [46]。 $\delta H\alpha(\text{exp})$ 是實驗勝肽上感興趣的殘基的化學位移。 $\delta H\alpha(\text{U})$ 是完全展開的參考勝肽上對應目標殘基的化學位移。 $\delta H\alpha(\text{F})$ 是完全折疊的參考勝肽上對應目標殘基的化學位移。

$$\Delta\delta H\alpha(\text{exp}) = \delta H\alpha(\text{exp}) - \delta H\alpha(\text{U}) \quad (1)$$

$$\Delta\delta H\alpha(\text{F}) = \delta H\alpha(\text{F}) - \delta H\alpha(\text{U}) \quad (2)$$

4.4. $^3J_{NH\alpha}$ Spin–Spin Coupling Constant

The peak-to-peak separation in the absorptive (v_a) and dispersive (v_d) DQF-COSY spectra was measured to derive the $^3J_{NH\alpha}$ coupling constant using the values along the f2 axis. The absorptive (v_a) and dispersive (v_d) values were used to derive the coupling constants from the square root of the single real root using Equation (3) [48].

$$J^6 - v_a^2 J^4 + \left(-\frac{9}{4} v_a^4 + \frac{3}{2} v_a^2 v_d^2 + \frac{3}{4} v_d^4 \right) J^2 + \frac{81}{64} v_a^6 - \frac{9}{16} v_a^4 v_d^2 - \frac{21}{32} v_a^2 v_d^4 - \frac{1}{16} v_d^6 - \frac{v_d^8}{64 v_a^2} = 0 \quad (3)$$

4.5. Interproton Distance Determination via NOE Integration

The NOE cross-peaks of all peptides were assigned from the corresponding ROESY spectra. Integration was performed based on a Gaussian peak model to obtain the intensity of cross-peaks. The distance between the two β -hydrogen atoms on the proline side chain (1.77 Å) was set as the standard to derive the interproton distance for the cross-peak of interest using Equation (4). The distances were grouped into short (≤ 2.5 Å), medium (2.5–3.5 Å), and long (> 3.5 Å) for the illustrations in the Wüthrich diagrams (Figures S39–S50).

$$R_{\text{residue}} = 1.77 \times 10^{-10} \times \left(\frac{I_{\text{standard}}}{I_{\text{residue}}} \right)^{\frac{1}{6}} \quad (4)$$

4.6. Fraction Folded Population and Folding Free Energy (ΔG_{fold})

The equilibrium constant between the unfolded and folded states of an experimental peptide is the ratio of the folded and unfolded populations. The fraction folded population for each residue was derived from the chemical shift data using Equation (5). The folding free energy ΔG_{fold} for each residue was derived using Equation (6). The fraction folded populations and folding free energy (ΔG_{fold}) of peptides were obtained by averaging the corresponding values for residues at positions 2, 3, 9, and 10 [3,19,22,25–27,38]. These four positions were in the middle of the strands, neither at the termini (which would suffer from the end fraying effect) nor directly attached to the turn (which would promote sheet formation). Furthermore, the average value of these four positions would provide equal representation for both hydrogen-bonded positions (residues 3 and 10) and non-hydrogen-bonded positions (residues 2 and 9), and also equal representation for both strands in the hairpin (i.e., the N-terminal strand: residues 2 and 3; the C-terminal strand: residues 9 and 10).

$$\text{Fraction Folded Population} = \frac{\delta H\alpha(\text{exp}) - \delta H\alpha(\text{U})}{\delta H\alpha(\text{F}) - \delta H\alpha(\text{U})} \times 100\% \quad (5)$$

$$\Delta G_{\text{fold}} = -RT \ln \frac{\delta H\alpha(\text{exp}) - \delta H\alpha(\text{U})}{\delta H\alpha(\text{F}) - \delta H\alpha(\text{exp})} \quad (6)$$

4.7. Double Mutant Cycle Analysis

Double mutant cycle analysis [50] was performed to determine the interaction free energy (ΔG_{int}) between charged residues Zbb2 and Xaa9 in HPDZbbXaa peptides using Equation (7). This analysis accounted for the effect of each charged residue (individually) on strand stability using data from the corresponding Ala-containing peptides HPDZbbAla and HPDAlaXaa [38] to determine the Zbb2-Xaa9 ion-pairing interaction exclusively. The peptide with Ala incorporated at positions 2 and 9, HPDAlaAla, was used as the reference peptide.

$$\Delta G_{\text{int}} = (\Delta G_{\text{HPDZbbXaa}} - \Delta G_{\text{HPDAlaAla}}) - (\Delta G_{\text{HPDZbbAla}} - \Delta G_{\text{HPDAlaAla}}) - (\Delta G_{\text{HPDAlaXaa}} - \Delta G_{\text{HPDAlaAla}}) \quad (7)$$

4.4. $^3J_{NH\alpha}$ Spin–Spin Coupling Constant

測量吸收 (ν_a) 和色散 (ν_d) DQF-COSY 光譜中的峰到峰分離，以使用沿 f2 軸的值導出 $^3J_{NH\alpha}$ 耦合常數。吸收 (ν_a) 和色散 (ν_d) 值用於使用方程式 (3) [48] 從單一實根的平方根導出耦合常數。

$$J^6 - \nu_a^2 J^4 + \left(-\frac{9}{4} \nu_a^4 + \frac{3}{2} \nu_a^2 \nu_d^2 + \frac{3}{4} \nu_d^4 \right) J^2 + \frac{81}{64} \nu_a^6 - \frac{9}{16} \nu_a^4 \nu_d^2 - \frac{21}{32} \nu_a^2 \nu_d^4 - \frac{1}{16} \nu_d^6 - \frac{\nu_d^8}{64 \nu_a^2} = 0 \quad (3)$$

4.5. Interproton Distance Determination via NOE Integration

所有勝肽的 NOE 交叉峰均根據相應的 ROESY 光譜進行分配。基於高斯峰模型進行積分以獲得交叉峰的強度。將脯氨酸側鏈上兩個 β -氫原子之間的距離 (1.77 Å) 設定為標準，以使用方程式 (4) 推導出感興趣的交叉峰的質子間距。在 Wüthrich 圖解中，距離分為短距離 (≤ 2.5 Å)、中距離 (2.5–3.5 Å) 和長距離 (> 3.5 Å) (圖 S39–S50)。

$$R_{\text{residue}} = 1.77 \times 10^{-10} \times \left(\frac{I_{\text{standard}}}{I_{\text{residue}}} \right)^{\frac{1}{6}} \quad (4)$$

4.6. Fraction Folded Population and Folding Free Energy (ΔG_{fold})

實驗勝肽的未折疊和折疊狀態之間的平衡常數是折疊和未折疊群體的比率。每個殘基的折疊群體分數是使用方程式 (5) 從化學位移資料導出的。每個殘基的折疊自由能 ΔG_{fold} 使用方程式 (6) 導出。勝肽的折疊群體分數和折疊自由能 (ΔG_{fold}) 透過對位置 2、3、9 和 10 處殘基的相應值進行平均來獲得 [3,19,22,25–27,38]。這四個位置位於股線的中間，既不在末端（會受到端部磨損效應的影響），也不直接連接到轉彎處（這會促進片材形成）。此外，這四個位置的平均值將為氫鍵位置（殘基 3 和 10）和非氫鍵位置（殘基 2 和 9）提供相同的表示，並且還為髮夾中的兩條鏈提供相同的表示（即，N 端鏈：殘基 2 和 3；C 端鏈：殘基 9 和 10）。

$$\text{Fraction Folded Population} = \frac{\delta H\alpha(\text{exp}) - \delta H\alpha(\text{U})}{\delta H\alpha(\text{F}) - \delta H\alpha(\text{U})} \times 100\% \quad (5)$$

$$\Delta G_{\text{fold}} = -RT \ln \frac{\delta H\alpha(\text{exp}) - \delta H\alpha(\text{U})}{\delta H\alpha(\text{F}) - \delta H\alpha(\text{exp})} \quad (6)$$

4.7. Double Mutant Cycle Analysis

進行雙突變體循環分析 [50]，使用方程式 (7) 確定 HPDZbbXaa 勝肽中帶電殘基 Zbb2 和 Xaa9 之間的相互作用自由能 (ΔG_{int})。該分析使用來自相應的含 Ala 的勝肽 HPDZb-bAla 和 HPDAlaXaa [38] 的數據來解釋每個帶電殘基（單獨）對鏈穩定性的影響，以專門確定 Zbb2-Xaa9 離子對相互作用。在位置 2 和 9 摻入 Ala 的勝肽 HPDAlaAla 用作參考勝肽。

$$\Delta G_{\text{int}} = (\Delta G_{\text{HPDZbbXaa}} - \Delta G_{\text{HPDAlaAla}}) - (\Delta G_{\text{HPDZbbAla}} - \Delta G_{\text{HPDAlaAla}}) - (\Delta G_{\text{HPDAlaXaa}} - \Delta G_{\text{HPDAlaAla}}) \quad (7)$$

5. Conclusions

The effects of charged amino acid side-chain length on diagonal cross-strand ion-pairing interaction between carboxylate- and ammonium-containing residues was investigated in a β -hairpin by NMR methods. HPDAadDap exhibited the highest fraction folded population because the long Aad side chain inherently favors strand formation coupled with the reasonably stabilizing diagonal Aad2-Dap9 ion-pairing interaction. Furthermore, the diagonal Asp2-Dab9 ion-pairing interaction was the most stabilizing interaction, most likely representing an optimal balance between the side-chain conformational entropic penalty to enable the ion-pairing interaction and the number of side-chain conformations that can accommodate the interaction. These results should be useful for stabilizing β -sheet containing molecular entities for various applications. β -Strand containing motifs have been used for specific recognition of various biomolecules including ATP [56], DNA [57,58], and RNA [59–64]. Furthermore, β -sheet containing peptides have been developed for ion channel blocking [65], bacterial endotoxin inhibition [66], antimicrobial activity [67], hydrogel formation [68–75], and intracellular delivery [76,77]. The results in this study may also facilitate the design of β -strand containing molecules for reducing amyloid [78], or for inhibiting protein–protein interactions involving β -structures [79,80].

Supplementary Materials: The following are available online at <https://www.mdpi.com/article/10.3390/molecules27134172/s1>, Tables S1–S36: The ^1H chemical shift assignments for the peptides, Tables S37–S45: The $^3J_{\text{NH}\alpha}$ values of the peptides, Figure S1: The $\text{H}\alpha$ chemical shift deviations for the residues in the experimental HPDZbbXaa peptides, Figure S2: The $\text{H}\alpha$ chemical shift deviations for the residues in the fully folded reference HPDFZbbXaa peptides, Figures S3–S38: The NOEs observed involving the side chains of the peptides, Figures S39–S50: Wüthrich diagrams of the backbone NOE connectivities involving the α -protons and amide protons for the peptides, Figure S51: The fraction folded of the residues in the peptides, Figure S52: The ΔG_{fold} of the residues in the peptides.

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5. 結論

透過 NMR 方法研究了 β -髮夾中帶電氨基酸側鏈長度對含羧酸根和含胺殘基之間對角交叉鏈離子對相互作用的影響。HPDAadDap 表現出最高分數的摺疊群體，因為長 Aad 側鏈本質上有利於鏈形成，再加上相當穩定的對角線 Aad2-Dap9 離子對相互作用。此外，對角線 Asp2-Dab9 離子對相互作用是最穩定的相互作用，很可能代表了實現離子對相互作用的側鏈構象熵罰分與可容納相互作用的側鏈構象數量之間的最佳平衡。這些結果對於穩定包含分子實體的 β -片層以用於各種應用應該是有用的。含有 β -鏈的基序已用於特異性識別各種生物分子，包括 ATP [56]、DNA [57,58] 和 RNA [59–64]。此外，含有 β -片層的勝肽已被開發用於離子通道阻斷[65]、細菌內毒素抑制[66]、抗菌活性[67]、水凝膠形成[68–75]和細胞內遞送[76,77]。這項研究的結果也可能有助於設計含有 β 鏈的分子，以減少澱粉樣蛋白 [78]，或抑制涉及 β 結構的蛋白質-蛋白質相互作用 [79,80]。

補充資料：以下內容可上網取得：<https://www.mdpi.com/article/10.3390/molecules27134172/s1>，表 S1–S36：勝肽的 ^1H 化學位移分配，表 S37–S45：勝肽的 $^3J_{\text{NH}\alpha}$ 值，圖 S1：勝肽的 HPD，HPDFxP，S2：完全折疊的參考 HPDFZbbXaa 肽中殘基的 $\text{H}\alpha$ 化學位移偏差，圖 S3–S38：觀察到的涉及肽側鏈的 NOE，圖 S39–S50：涉及 α -質子和酰胺質子的骨架 NOE 連接的 Wüthrich 圖肽，圖 S51：勝肽中殘基的摺疊分數，圖 S52：勝肽中殘基的 ΔG_{fold} 。

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